

● MEDIUM

● ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

06

6 questions · ~9 min

Q1

ADVANCED MATH

$$y = ax^2 - c$$

In the equation above, a and c are positive constants. How many times does the graph of the equation above intersect the graph of the equation $y = a + c$ in the xy -plane?

- A. Zero
- B. One
- C. Two
- D. More than two

Q2

GRID-IN

ADVANCED MATH

$$7x^2 - 20x - 32 = 0$$

What is the positive solution to the given equation?

STUDENT-PRODUCED RESPONSE

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.	/	.	/	.
0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ADVANCED MATH

$$3xx - 4x + 5 = 0$$

What is one of the solutions to the given equation?

- A. -4
- B. 0
- C. 3
- D. 5

Q4

ADVANCED MATH

$$52x^3 + 64x^4 - 81 = 0$$

How many distinct real solutions does the given equation have?

- A. Exactly two
- B. Exactly three
- C. Exactly five
- D. Exactly seven

Q5

GRID-IN

ADVANCED MATH

$$y = x^2 - 4x + 4$$

$$y = 4 - x$$

If the ordered pair (x, y) satisfies the system of equations above, what is one possible value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$7m = 5n + p$$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives n in terms of m and p ?

A. $n = \frac{5p}{7m}$

B. $n = \frac{7m}{5} - p$

C. $n = 57m + p$

D. $n = 7m - 5 - p$